



Review

Executive function measures of participants with mild cognitive impairment: Systematic review and meta-analysis of event-related potential studies

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ABSTRACT

Objective: Objective measurements of executive functions using event-related potential (ERP) may be used as markers for differentiating healthy controls (HC) from patients with mild cognitive impairment (MCI). ERP is non-invasive, cost-effective, and affordable. Older adults with MCI demonstrate deteriorated executive function, serving as a potentially valid neurophysiological marker for identifying MCI. We aimed to review published ERP studies on executive function in older adults with MCI and summarize the performance differences by component between healthy older adults and older adults with MCI.

Methods: Eight electronic databases (Web of Science, PubMed, ScienceDirect, American Psychological Association PsycNet, Cochrane Library, Scopus, Embase, and Ovid) were searched for the study. Articles published from January 1 to December 31, 2022, were considered for this review. A random-effects meta-analysis and between-study heterogeneity analysis were conducted using Comprehensive Meta-Analysis V3.0 software.

Results: We identified 7829 articles of which 28 met the full inclusion criteria and were included in the systematic review and analyses. Our pooled analysis suggested that participants with MCI can be differentiated from HC by significant P200, P300, and N200 latencies. The P100 and P300 amplitudes were significantly smaller in participants with MCI when compared with those in the HCs, and the patients with MCI showed increased N200 amplitudes. Our findings provide new insights into potential electrophysiological biomarkers for diagnosing MCI.

1. Introduction

Dementia is a chronic disease characterized by a progressive decline in cognitive function that places a significant strain on individuals and society. The incidence of Alzheimer's Disease (AD) among older adults aged >65 years increases by one fold every 5 years, and people >85 years old have a 30 % chance of developing the disease (Zhao et al., 2021). The mortality rate of this disease is also increasing every year in developed countries, such as the United States (Taylor et al., 2017). In population-based epidemiological studies, the prevalence of mild

cognitive impairment (MCI) in adults over 65 years of age ranges from 3 to 19 %, and over time, some individuals appear to stabilize or return to normal, but more than half of those affected develop dementia within 5 years (Gauthier et al., 2006). An estimated 6.7 million Americans aged 65 and older currently have AD. This number could increase to 13.8 million by 2060 if there are no medical breakthroughs that prevent, slow, or cure AD (Association, A. P., 2013). If patients with MCI have mild neuronal damage, the disease may be reversible to a certain extent, providing an excellent window for AD prevention and treatment. Therefore, identifying neuroelectrophysiological markers for effective

Abbreviations: MCI, mild cognitive impairment; HC, healthy control; ERP, event-related potential; AD, Alzheimer's Disease; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; EEG, event-related electroencephalography; JBI PAGES, Joanna Briggs Institute Practical Application of Clinical Evidence System; CMA, Comprehensive Meta-Analysis.

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identification of MCI could delay disease progression, facilitate early intervention, and prevent the development of AD (Ward et al., 2021).

No targeted diagnostic criteria for MCI are included in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (Association, A. P., 2013); however, phenomenological descriptions of the cluster of symptoms associated with dementia are described. This method of diagnosis is disadvantageous owing to strong subjectivity, influence of patient education, and insensitivity in the early stage (Cassani et al., 2018). Among the existing MCI diagnostic methods, the Montreal Cognitive Assessment (MoCA), Mini-Mental State Examination (MMSE), and other comprehensive neural tests have shown varied specificity and sensitivity, with some individuals meeting screening criteria for cognitive impairment on one test but not the other (Werner and Karczyn, 2008). Furthermore, even comprehensive neuropsychological measures are not specific or sensitive enough to detect very early cognitive deficits in patients with MCI (Werner and Karczyn, 2008).

Biomarkers for MCI have been identified using a number of techniques such as positron emission tomography, magnetic resonance imaging, and lumbar puncture for cerebrospinal fluid. However, these techniques are difficult to perform in low- and middle-income countries, and also remote areas of high-income countries, where the lack of expertise and facilities required for these tests limits their application (Khouri and Ghossoub, 2019).

Measuring electrical activity in the brain using event-related electroencephalography (EEG) is a cost-effective method for detecting the early brain changes associated with AD and its prodromal MCI phase (Babiloni et al., 2013). Combining EEG with a task paradigm elicits neural responses to specific sensory, cognitive, and motor processes. These responses are also known as event-related potentials (ERPs). An ERP is the result of the brain adjusting from a disconnected (high entropy) to an ordered (low entropy) state in response to a specific task (Cek et al., 2010). Hence, ERPs reflect changes in the state of electrical activity in the brain in response to different tasks, and are a non-invasive tool with high temporal resolution and advantages for quantifying the sequence of different cognitive processing stages. Moreover, ERP is an objective tool for assessing the cognitive state and other brain functions (Takeda and Kakigi, 2010). Some ERP components of MCI are sensitive to cognitive decline, and changes in ERP indicators precede changes in behavior (Chiang et al., 2018; Mudar et al., 2016; Ramos-Goicoa et al., 2016). ERP components are likely to be early electrophysiological diagnostic biomarkers for MCI (Yamasaki et al., 2016). Because of the high temporal resolution of EEG, it can sensitively capture the neural activity characteristics behind the executive functional tasks of patients with MCI. Therefore, the construction of an objective index system for MCI based on an event-related EEG can provide new technical means for its early identification.

Some researchers have found that patients with MCI with executive function impairment have faster rates of disease progression and increased risk of progression to AD (Cid et al., 2019; Gu and Zhang, 2017; Junquera et al., 2020). Executive function is a series of higher-order cognitive functions required in the process of setting goals, and making and implementing plans (Lacreuse et al., 2020). Diamond proposed a multidimensional structural model of executive function comprised of three core components: inhibitory control, working memory, and cognitive flexibility (Diamond, 2012). It has been suggested that executive function processes (e.g., inhibitory deficits, working memory deficits) may be present in the preclinical phase of MCI (Adlam et al., 2006; Bélanger and Belleville, 2009; Chiang et al., 2015; Price et al., 2012). Impaired executive function, such as the inability to inhibit irrelevant information from entering the brain and processing working memory lists, may lead to difficulties in episodic memory retrieval (Chehrehnegar et al., 2022), impaired ability to deal with problems (Lee et al., 2019), and decreased motor flexibility and language ability (Oyama et al., 2019). As an example, based on Baddeley's working memory theory (Baddeley, 1992), the inhibitory control (which is a subcomponent of executive function) deficits that do not effectively

inhibit irrelevant information from entering the working memory model will cause or exacerbate memory deficits in patients with MCI. Consequently, this study focuses on executive function that may be useful for identification of neurophysiological markers enabling early recognition of MCI in older adults.

As mentioned earlier, executive function is divided into three parts, each corresponding to a number of paradigms. The inhibitory control inclusion paradigm includes the Stroop, flanker, Go-NoGo, Simon, and stop-signal tasks. Cognitive flexibility involves paradigms such as the Wisconsin Card Sorting Test (WCST). The working memory paradigm encompasses the N-back task, task switching, delayed matching-to-sample (DMS) task, and the auditory/visual oddball paradigm (West et al., 2010). In this study, we retrieved all the above-mentioned executive function paradigms.

The more common EEG components are the MMN, N100, N200, P100, P200, and P300 amplitudes and latencies. MMN is obtained by deducting the ERP elicited by standard stimuli from that elicited by deviant stimuli during the oddball task. It is a negative component appearing approximately 160–200 ms after stimulus onset, and is likely to be produced by an autonomic novelty-sensing process rather than attention to stimuli (Picton et al., 2000). The N100 is composed of components derived from the auditory cortex and others (Wolpaw and Penry, 1975). The N200, a negative component that appears 180–350 ms after stimulus onset, is related to selective pre-attentive stimulus evaluation and discrimination (Bennys et al., 2007). The P100 in the occipital area is related to the global processing of visual perception (Liu et al., 2002). The P200 is traditionally known as an automatically produced exogenous response, although its amplitude and latency might change with selective attention (Crowley and Colrain, 2004). The P300 is a positive component peaking between 300 and 500 ms after stimulus onset. The P300a is mainly distributed in the frontal electrodes and is related to automatic attention processing. The P300b, the classical P300, is elicited by working memory updates, and is distributed primarily in parietal electrodes (Polich, 2007). These paradigms, components, and indicators can reflect the differences between patients with MCI and healthy groups of people, to a certain extent.

Executive function-related ERPs have been insufficiently utilized for diagnosing MCI. Furthermore, comparative analyses of these components between patients with MCI and healthy controls (HC) have yielded inconsistent results across investigations. Cid-Fernández et al. noted that P200 latency was delayed in patients with MCI when compared with HCs (Cid-Fernández et al., 2017). However, López et al. found no significant differences between the two groups of subjects in the P200 component, but identified significant differences in the P300 component (López Zunini et al., 2016a, 2016b). Questions persist regarding factors causing discrepancies in different study results and on the general trends in existing research on the various components of executive function, highlighting the urgency of a meta-analysis of these components in patients with MCI. These questions suggest that a meta-analysis of these components in patients with MCI is needed. This study aimed to determine: 1) the magnitude of executive function-related ERP latency and amplitude abnormalities present in patients with MCI compared with those in HCs, 2) whether executive function-related ERPs can differentiate between the two groups (MCI versus HC), and 3) executive function-related ERPs as potential biomarkers for pre-clinical and early clinical stages of MCI. Our findings will provide a new theoretical and practical basis for early diagnosis and prevention of MCI.

2. Materials and methods

2.1. Protocol and registration

This review was conducted in accordance with the 2020 PRIMSA guidelines (Page et al., 2021), and the protocol was registered with PROSPERO (CRD42023432579).

2.2. Eligibility criteria

The inclusion criteria were as follows:

- Population: MCI group (people aged >60 years with MCI), HC group (people aged >60 years without MCI);
- Subject: EEG study of the experimental paradigm of executive function in people with MCI based on ERP;
- Comparator: healthy people age-matched with the control group;
- Outcome: peak values and latencies of ERP components between HCs and patients with MCI are clearly reported along with the sample size;
- Study type: all types of quantitative empirical studies are included.

To assess the most current technology for diagnosing MCI (specifically EEG characteristics related to executive function in older adults) as comprehensively as possible, we considered studies published from January–December 2022. To conduct a scientific meta-analysis of the evidence that might contribute to a better understanding of the role of electrophysiological signals in this field, we developed the following exclusion criteria:

- Population: people aged <60 years;
- Subject: no mention of executive function or did not evaluate peak and latent EEG values;
- Outcome: peak values and latencies of ERP components between HCs and patients with MCI are not clearly reported along with the sample size;
- Study type: qualitative intervention and observational studies.

Only studies published in English were included. To ensure that our study was based on reliable data, we only included published and peer-reviewed studies and excluded gray literature. Moreover, we excluded studies with repeated data. If studies with repeated data were encountered, only one was included, and the study with the largest sample size and the most complete data was selected for the next stage of analysis.

2.3. Study identification and selection

Electronic searches for this systematic review and meta-analysis were conducted in January 2023 using eight major electronic databases (Web of Science, PubMed, ScienceDirect, American Psychological Association PsycNet, Cochrane Library, Scopus, Embase, and Ovid). Some examples of the keywords that we adopted for the first group of elements (i.e., cognitive component) include: “executive function,” “executive dysfunction,” “inhibitory control,” “conflict control,” “response inhibition,” “cognitive flexibility,” and “working memory.” Some examples of the keywords used for the second group (i.e., target population) include: “mild cognitive impairment,” “MCI,” “cognitive impairment,” “memory impairment,” “prodromal Alzheimer’s Disease,” “cognitive decline,” “cognitive ability,” “cognitive development,” and “cognitive improvement.” Examples of the keywords used for the third group (i.e., outcome) include: “event-related potential,” “ERP,” “electroencephalography,” and “EEG.” The first author performed the database searches and removal of duplicates. Screening of titles and abstracts and full-text evaluations were independently conducted by the first and second authors without the use of automated tools. Cases of unclear eligibility were assessed and resolved through group discussions with the remaining authors.

2.4. Domains

We focused on the executive functions domain according to the following experimental paradigms: Go/No-Go, N-back, Simon, flanker, and DMS tasks; task switching; and the auditory/visual oddball paradigm. The standardized mean difference was used to measure the effects.

We aimed to quantify the difference between the MCI and HC groups.

2.5. Data extraction

Data extraction was performed independently by the first and second authors, and the interval between two encodings was 2 weeks. The consistency coefficient between the two encodings was 93.8 %. Discrepancies were resolved via group discussions with the other authors.

2.6. Risk of bias

We applied Begg’s test and used funnel plots to assess the risk of publication bias. Both publication bias analyses were performed using Comprehensive Meta-Analysis (CMA) V3.0 software.

2.7. Evaluation of the literature

In addition to rating the risk of bias for each study, we used the Joanna Briggs Institute Practical Application of Clinical Evidence System (JBI PACES) to evaluate the overall quality of the evidence across studies (Sianes-Gallén et al., 2021). Studies were scored on a scale of 1–20, with higher scores indicating higher qualities of evidence. The first and second authors conducted independent evaluations of the literature and disagreements were resolved via group discussion with the other authors.

2.8. Data collection

Finally, 28 ERP studies were reviewed. The following data were recorded from the full texts of these ERP studies: author(s), publication year, participant demographics (sample size, age, sex, education level, and MMSE score), experimental task, ERP component and its window, and study results.

2.9. Meta-analysis of ERP studies

The mean, standard deviation, and 95 % confidence intervals were calculated as the effect size of each ERP component explored by ≥ 3 ERP studies. An effect size of 0.2–0.5 was considered a small effect, 0.5–0.8 was a medium effect, and ≥ 0.8 was a large effect (Sullivan and Feinn, 2012). The Q test was used to estimate heterogeneity between studies, and I^2 was computed to assess the degree of variation derived from heterogeneity. If the heterogeneity of these studies was invariably high (Q test, $p \leq 0.05$), we generated a random-effects model summary effect size. Inversely, fixed-effects models were applied to summarize the effect size in the absence of between-study heterogeneity. Therefore, this study used a random-effects model. In reference to the previous study (Gu and Zhang, 2017), our analyses were based on the ERP components only. All statistical analyses were performed using the CMA V3.0 software.

To explore the reasons for heterogeneity, we conducted meta-regression analyses with sex ratio, year of education, and age as moderating variables, and subgroup analyses with paradigms as subgroups.

3. Results

3.1. Search results

Fig. 1 shows the initial search strategy and gradual exclusion procedures. A total of seven sets of titles and abstracts were screened against the eligibility criteria, with six sets remaining after removing duplicates. Among the screened articles, six were excluded based on the title and abstract and nine were selected for full-text review. A total of 28 articles met the full inclusion criteria and were included in the various meta-analyses.

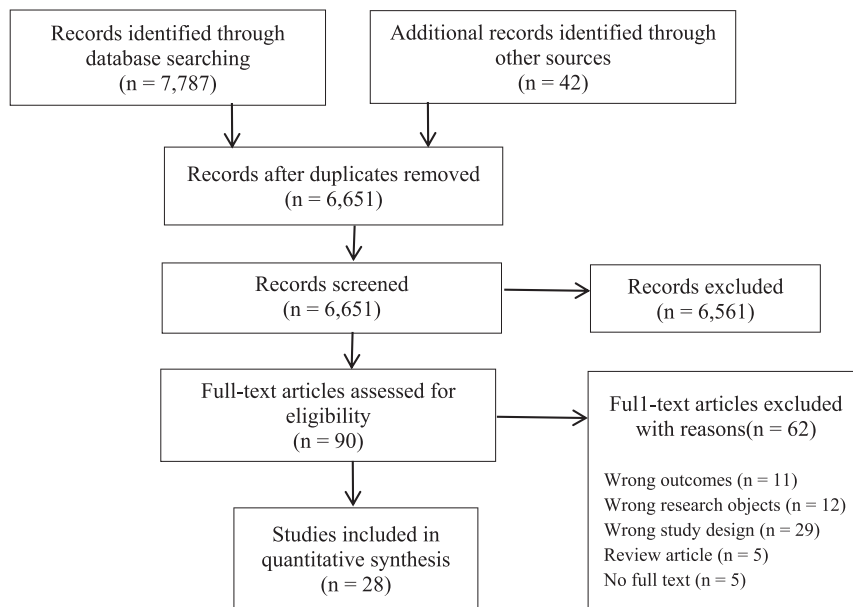


Fig. 1. PRISMA flow diagram of search result.

3.2. Study characteristics

The study characteristics and results are summarized in Table 1. The characteristics of the 28 ERP studies on patients with MCI are summarized as follows: studies that compared MMN and P100 amplitudes and latencies between HC and MCI groups ($n = 3$), studies exploring MCI-related alterations of the N100 component ($n = 5$), studies that evaluated MCI-related alterations of P200 amplitudes and latencies ($n = 5$ and $n = 6$, respectively), studies that evaluated MCI-related N200 amplitude and latency changes ($n = 15$ and $n = 14$, respectively), and studies that assessed MCI-related P300 amplitude and latency alterations ($n = 19$ and $n = 15$, respectively).

3.3. Risk of bias

We assessed the risk of bias for each outcome. Begg's test and funnel plots revealed no significant publication bias for the MMN, N200, P100, and P200 amplitude components and MMN, P100, P200, and P300 latency components; whereas publication bias was observed for the N100 and P300 amplitude components and the N200 latency components. Funnel plots are shown in the Supplementary Material.

3.4. Heterogeneity

The percentage of variation across studies (I^2 statistic) owing to heterogeneity, as well as Cochran's Q-value and significance level, are presented in the forest plot for each meta-analysis (Figs. 2–3). An I^2 value $\leq 25\%$ was considered low heterogeneity, between 26 and 50 % was moderate heterogeneity, and $\geq 75\%$ was substantial heterogeneity (Jiang et al., 2017). A higher percentage of variation (I^2) across studies is indicative of greater variations in study outcomes or between study variations in clinical heterogeneity (Riekkinen Jr. et al., 1997). Despite efforts to reduce clinical heterogeneity using strict inclusion criteria and only pooling studies with similar designs, significant heterogeneity was still observed across some studies.

The following values were observed: P200 amplitude ($I^2 = 72.44\%$, $p = 0.000$), P200 latency comparing MCI to controls only ($I^2 = 32.54\%$, $p = 0.122$), N100, N200 amplitude ($I^2 > 75\%$, $p = 0.000$), MMN amplitude ($I^2 = 64.99\%$, $p = 0.009$), MMN latency comparing MCI to controls only ($I^2 = 46.21\%$, $p = 0.08$), N200, P300 amplitude ($I^2 < 80\%$, $p = 0.000$). This study detected significant heterogeneities between ERP-

related studies for MMN, N100, N200, P100, P200, and P300 amplitudes and MMN, N200, P100, P200, and P300 latencies. Notably, the P100 component heterogeneity analyses results were insignificant (amplitude: $I^2 = 0\%$, $p = 0.607$; latency: $I^2 = 0\%$, $p = 0.670$).

Controlling for factors such as disease severity and duration could reduce between-study heterogeneity (Tarawneh et al., 2021); however, achieving this can be difficult as studies reporting on MCI severity are limited that meta-analysis by severity classification is not possible. Moreover, even fewer studies have reported on the duration of MCI. As higher heterogeneity reduces confidence in the fixed effect, we applied a random-effects model in this study.

3.5. Meta-analysis of ERP studies

Our meta-analysis results showed reduced N200, P100, and P300 (Fig. 2C, D, F) amplitudes in patients with MCI compared with those in HCs (N200 is a negative wave; therefore, the amplitude result is taken as an absolute value). No significant differences in MMN, N100, and P200 amplitudes (Fig. 2A, B, E) were observed between patients with MCI and HCs. Moreover, our findings revealed prolonged N200, P200, and P300 latencies (Fig. 3B, D, E) in patients with MCI compared with those in HCs. Furthermore, no significant differences in MMN and P100 latencies (Fig. 3A, C) were observed between patients with MCI and HCs.

We performed a meta-regression analysis of the components that differed significantly from each other. We analyzed sex ratio, educational attainment, and mean age as moderating variables and the following results were observed. The N200 amplitude results for sex ratio ($R^2 = 0.637$, $F = 11.48$, $p < 0.01$), educational attainment ($R^2 = 0.037$, $F = 4.95$, $p < 0.05$), and mean age ($R^2 = 0.007$, $F = 6.21$, $p < 0.05$) were significant. The P300 amplitude results for sex ratio ($R^2 = -0.339$, $F = 71.49$, $p < 0.01$), educational attainment ($R^2 = -0.037$, $F = 95.48$, $p < 0.01$), and mean age ($R^2 = -0.006$, $F = 79.43$, $p < 0.01$) were significant. The N200 latency results for sex ratio ($R^2 = 0.351$, $F = 10.64$, $p < 0.01$), educational attainment ($R^2 = 0.026$, $F = 12.00$, $p < 0.01$), and mean age ($R^2 = 0.005$, $F = 20.93$, $p < 0.01$) were significant. The P200 latency results for sex ratio ($R^2 = 0.308$, $F = 7.16$, $p < 0.05$), educational attainment ($R^2 = 0.038$, $F = 9.68$, $p < 0.01$), and mean age ($R^2 = 0.008$, $F = 29.80$, $p < 0.01$) were significant. Lastly, the P300 latency results for sex ratio ($R^2 = 0.288$, $F = 6.09$, $p < 0.05$), educational attainment ($R^2 = 0.026$, $F = 10.67$, $p < 0.01$), and mean age ($R^2 = 0.004$, $F = 8.86$, $p < 0.01$) were significant.

Table 1

Characteristics of ERP studies included in the meta-analysis of MCI.

Study	n	Age (SD)	Sex (M/ F)	Education (SD)	MMSE (SD)	Task/ Paradigm	ERP component and window (in ms post- event)	Results		JBI PACES
								Amplitude	Latency	
(Mudar et al., 2016) (Mudar et al., 2016)	MCI	68.50	9/16	16.00	28.40	Go-NoGo	N200 (Fz, FCz, Cz) 150–300 ms; P300 (FCz, Cz, CPz) 250–650 ms.	N200(SiC-Go): MCI	N200(SiC-Go):	17
	25	(8.00)		(1.90)	(1.30)			> HC;	MCI > HC;	
	HC	65.40	9/16	16.60	28.60			N200(SiC-NoGo):	N200(SiC-NoGo):	
	25	(7.10)		(1.70)	(0.50)			MCI > HC;	MCI < HC;	
								N200(ObA-Go): MCI	N200(ObA-Go):	
								> HC;	MCI > HC;	
								N200(ObA-NoGo):	N200(ObA-NoGo):	
								MCI < HC;	MCI > HC;	
								P300(SiC-Go): MCI	P300(SiC-Go):	
								< HC;	MCI > HC;	
								P300(SiC-NoGo):	P300(SiC-NoGo):	
								MCI < HC;	MCI > HC;	
								P300(ObA-Go): MCI	P300(ObA-Go):	
								> HC;	MCI > HC;	
								P300(ObA-NoGo):	P300(ObA-NoGo):	
								MCI < HC.	MCI > HC.	
(López Zunini et al., 2016a)(López Zunini et al., 2016a)	MCI	75.60	7/8	14.67	/	Go-NoGo	N200 (Fz,FCz,Cz, CPz,Pz) 200–350 ms; P300 (Fz,FCz,Cz, CPz,Pz) 400–500 ms.	N200(Fz-NoGo):	N200(Fz-NoGo):	17
	15	(6.02)		(2.79)	/			MCI < HC;	MCI > HC;	
	HC	72.40	6/11	15.58	/			N200(FCz-NoGo):	N200(FCz-NoGo):	
	17	(15.87)		(3.04)				MCI < HC;	MCI > HC;	
								N200(Cz-NoGo):	N200(Cz-NoGo):	
								MCI < HC;	MCI > HC;	
								N200(CPz-NoGo):	N200(CPz-NoGo):	
								MCI > HC;	MCI > HC;	
								N200(Pz-NoGo):	N200(Pz-NoGo):	
								MCI > HC;	MCI > HC.	
								P300(Fz-Go): MCI <		
								HC;		
								P300(FCz-Go): MCI		
								< HC;		
								P300(Cz-Go): MCI <		
								HC;		
								P300(CPz-Go): MCI		
								< HC;		
								P300(Pz-Go): MCI <		
								HC;		
								P300(Fz-NoGo):		
								MCI < HC;		
								P300(FCz-NoGo):		
								MCI < HC;		
								P300(Cz-NoGo):		
								MCI < HC;		
								P300(CPz-NoGo):		
								MCI < HC;		
								P300(Pz-NoGo):		
								MCI < HC.		
(Cid-Fernández et al., 2017)(Cid-Fernández et al., 2017)	HC	67.00	6/13	9.80 (5.20)	28.00	Go-NoGo	N200 (Fz,Cz,Pz): Go (250–430 ms); NoGo (200–360 ms); P300 (Fz,Cz,Pz): Go (350–700 ms); NoGo (400–550 ms); PSW.	N200(Fz-Go): MCI >	N200(Fz-Go): MCI	18
	20	(9.80)			(1.50)			HC;	> HC;	
	MCI	69.90	15/	9.07 (4.42)	25.67			N200(Cz-Go): MCI	N200(Cz-Go): MCI	
	34	(9.14)	18		(2.53)			> HC;	> HC;	
								N200(Pz-Go): MCI >	N200(Pz-Go): MCI	
								HC;	> HC;	
								N200(Fz-NoGo):	N200(Fz-NoGo):	
								MCI > HC;	MCI > HC;	
								N200(Cz-NoGo):	N200(Cz-NoGo):	
								MCI > HC;	MCI > HC;	
								N200(Pz-NoGo):	N200(Pz-NoGo):	
								MCI > HC;	MCI > HC;	
								P300(Fz-Go): MCI >	P300(Fz-Go): MCI	
								HC;	> HC;	
								P300(Cz-Go): MCI >	P300(Cz-Go): MCI	
								HC;	> HC;	
								P300(Pz-Go): MCI >	P300(Pz-Go): MCI	
								HC;	> HC;	
								P300(Fz-NoGo):	P300(Fz-NoGo):	
								MCI > HC;	MCI > HC;	
								P300(Cz-NoGo):	P300(Cz-NoGo):	
								MCI > HC;	MCI > HC;	
								P300(Pz-NoGo):	P300(Pz-NoGo):	
								MCI > HC.	MCI > HC.	
(Cid et al., 2019)(Cid et al., 2019)	MCI	77.1	6/11	8.5 (4.50)	25.0	Go-NoGo	N200 (Cz,Pz): Go (250–430 ms);	N200(Cz-Go): MCI	N200(Cz-Go): MCI	17
	17	(5.40)			(2.80)			> HC;	> HC;	

(continued on next page)

Table 1 (continued)

Study	n	Age (SD)	Sex (M/ F)	Education (SD)	MMSE (SD)	Task/ Paradigm	ERP component and window (in ms post- event)	Results		JBI PACES
								Amplitude	Latency	
	HC 18	76.4 (4.00)	5/13	8.6 (4.60)	27.6 (2.10)		NoGo (200–360 ms); P300 (Cz,Pz): Go (450–750 ms); NoGo (400–650 ms).	N200(Cz-NoGo): MCI > HC; P300(Pz-Go): MCI < HC; P300(Cz-NoGo): MCI > HC.	N200(Cz-NoGo): MCI < HC; P300(Pz-Go): MCI < HC; P300(Cz-NoGo): MCI < HC.	
(Cid-Fernández et al., 2014)(Cid- Fernández et al., 2014)	MCI 30 HC 63	69.5 (8.20) 65.9 (8.00)	14/ 16 22/ 41	9.7 (4.30) 8.9 (4.80)	25.9 (2.40) 28.2 (1.50)	Go-NoGo	N200 (Cz): Go (200–300 ms); NoGo (200–350 ms).	N200(Cz-go): MCI < HC; N200(Cz-nogo): MCI < HC.	/	17
(Gu et al., 2019)(L. Gu et al., 2019)	MCI 39 HC 46	71.28 (5.98) 70.17 (5.62)	29/ 30 25/ 21	10.81 (2.93) 12.22 (2.96)	27.08 (2.11) 28.43 (1.30)	Go-NoGo; N-back	N200 (Fz,Cz,P1,Pz, P2) 150–250 ms; P300 (Fz,Cz,P1,Pz, P2) 250–450 ms.	N200(Fz-NoGo): HC < MCI; N200(Cz-NoGo): HC < MCI; N200(Pz-NoGo): HC < MCI; N200(Fz-Go): HC < MCI; N200(Cz-Go): HC < MCI; N200(Pz-Go): HC < MCI; P300(P1–0-back): MCI < HC; P300(Pz-0-back): HC > MCI; P300(P2–0-back): HC > MCI; P300(P1–1-back): HC > MCI; P300(Pz-1-back): HC > MCI; P300(P2–1-back): HC > MCI.	/	18
(Gu et al., 2017)(L. H. Gu et al., 2017)	MCI 39 HC 43	71.28 (5.98) 70.16 (5.43)	25/ 14 20/ 23	10.81 (2.93) 12.28 (2.97)	27.08 (2.11) 28.42 (1.33)	N-back	P300 (CP1,CPz,CP2, P1,Pz,P2) 300–500 ms.	P300(CP1–0-back): HC > MCI; P300(CPz-0-back): HC > MCI; P300(CP2–0-back): HC > MCI; P300(P1–0-back): HC > MCI; P300(P2–0-back): HC > MCI; P300(Cz-0-back): HC > MCI; P300(P2–0-back): HC > MCI; P300(CP1–1-back): HC > MCI; P300(CPz-1-back): HC > MCI; P300(CP2–1-back): HC > MCI; P300(P1–1-back): HC > MCI. P300(Cz-1-back): HC > MCI; P300(P2–1-back): HC > MCI.	/	18
(López Zunini et al., 2016b)(López Zunini et al., 2016b)	MCI 15 HC 17	75.60 (6.02) 72.41 (5.87)	7/8 6/11	14.67 (2.79) 15.59 (3.04)	/	N-back	P200 (Fz,FCz,Cz, CPz,Pz) 100–220 ms; N200 (Fz,FCz,Cz, CPz,Pz) 200–380 ms; P300 (Fz,FCz,Cz, CPz,Pz)400–500 ms.	P200(Fz): MCI < HC; N200(Fz): MCI > HC; P300(Fz): MCI < HC; P200(FCz): MCI < HC; N200(FCz): MCI < HC; P300(FCz): MCI < HC; P200(Cz): MCI < HC; N200(Cz): MCI < HC; P300(Cz): MCI < HC; P200(CPz): MCI < HC; P300(CPz): MCI < HC;	P200(Fz): MCI > HC; N200(Fz): MCI > HC; P200(FCz): MCI > HC; N200(FCz): MCI > HC; P200(Cz): MCI > HC; N200(Cz): MCI > HC; P200(CPz): MCI > HC; N200(CPz): MCI > HC;	17

(continued on next page)

Table 1 (continued)

Study	n	Age (SD)	Sex (M/F)	Education (SD)	MMSE (SD)	Task/Paradigm	ERP component and window (in ms post-event)	Results		JBI PAGES
								Amplitude	Latency	
(Missonnier et al., 2007)(Missonnier et al., 2007)	MCI	82.08 (7.02)	6/23	1.47 (0.72)	26.01 (1.90)	N-back	P100 (O1, Oz, O2, P3, Pz, P4); N160 (O1, Oz, O2, P3, Pz, P4); P200 (F3, Fz, F4, C3, Cz, C4); N200 (F3, Fz, F4, C3, Cz, C4); ERS (P3, Pz, P4) 1000–1700 ms.	N200(CPz): MCI < HC; P300(CPz): MCI < HC; P200(Pz): MCI < HC; N200(Pz): MCI > HC; P300(Pz): MCI < HC.	P200(Pz): MCI > HC; N200(Pz): MCI > HC.	17
	29									
	16	71.63 (8.99)	6/10	2.20 (0.79)	28.57 (1.16)					
(Mowszowski et al., 2012)(Mowszowski et al., 2012)	MCI	67.30 (8.10)	13/15	13.30 (3.50)	27.9 (1.60)	Auditory Odd-ball Paradigm	MMN (Fz, Cz, M1, M2) 100–250 ms.	MMN(Cz): MCI > HC; MMN(Fz): MCI < HC; MMN(M1): MCI < HC; MMN(M2): MCI < HC.	MMN(Cz): MCI < HC; MMN(Fz): MCI < HC; MMN(M1): MCI > HC; MMN(M2): MCI > HC.	18
	28									
	14	64.90 (4.00)	5/9	13.40 (2.80)	29.10 (1.00)					
(van Deursen et al., 2009)(van Deursen et al., 2009)	MCI	70.60 (7.20)	12/8	/	26.30 (1.60)	Auditory Odd-ball Paradigm	N200 (Cz) 200–250 ms; P300 (Fz, Cz, Pz) 300–600 ms.	P300(Pz): MCI < HC; N200(Cz): HC < MCI	P300(Pz): HC < MCI; N200(Cz): HC < MCI	15
	20									
	20	69.50 (6.10)	12/8	/	29.3 (0.80)					
(Frodl et al., 2002)(Frodl et al., 2002)	MCI	66.20 (11.30)	10/16	/	27.5 (1.60)	Auditory Odd-ball Paradigm	P300 (F3, F4, P3, P4) 240–366 ms.	P300(TB): HC < MCI; P300(TS): MCI < HC	P300(TB): HC < MCI; P300(TS): HC < MCI	15
	26									
	26	64.90 (10.90)	16/27	/	29.70 (0.50)					
(Golob et al., 2002)(Golob et al., 2002)	MCI	76.50 (2.70)	11/4	16.10 (2.40)	27.70 (1.90)	Auditory Odd-ball Paradigm	P50 (Cz) 40–80 ms; N100 (Cz) 80–160 ms; P200 (Cz) 150–250 ms; N200 (Cz) 175–250 ms; P300 (Pz) 250–600 ms.	P50(Cz): MCI > HC; N100(Cz): MCI < HC; P200(Cz): MCI > HC.	P50(Cz): MCI > HC; N100(Cz): MCI > HC; P200(Cz): MCI > HC.	15
	15									
	12	72.80 (7.80)	3/9	16.70 (2.50)	29.20 (0.80)					
(Irimajiri et al., 2005)(Irimajiri et al., 2005)	MCI	/	/	/	27.50 (1.70)	Auditory Odd-ball Paradigm	P50 (Cz) 25–80 ms; Nb (Cz) 60–130 ms; Pa (Cz) 120–245 ms.	/	Pa(Cz): MCI > HC; Nb(Cz): MCI > HC; P50(Cz): MCI > HC.	13
	16									
	15	/	/	/	29.30 (0.80)					
(Lai et al., 2010)(Lai et al., 2010)	MCI	68.00 (8.70)	11/7	9.90 (5.20)	23.07 (0.84)	Auditory Odd-ball Paradigm	N100 (Cz) 75–150 ms; P200 (Cz) 75–150 ms; N200 (Cz) 190–360 ms; P300 (Fz, Cz, Pz) 250–500 ms.	N100(Cz): MCI < HC; P200(Cz): MCI > HC; N200(Cz): MCI > HC; P300(Fz): MCI > HC; P300(Cz): MCI > HC; P300(Pz): MCI < HC.	N100(Cz): MCI < HC; P200(Cz): MCI < HC; N200(Cz): MCI > HC; P300(Fz): MCI > HC; P300(Cz): MCI > HC; P300(Pz): MCI > HC.	17
	18									
	14	64.80 (7.80)	9/5	9.70 (4.16)	28.25 (1.52)					
(Lister et al., 2016)(Lister et al., 2016)	MCI	76.80 (4.90)	9/4	15.00 (2.60)	/	Auditory Odd-ball Paradigm	P50 (FCz) 19–69 ms; N100 (FCz) 68–118 ms; P200 (FCz) 188–238 ms.	P100(FCz-pure tone stimulus): MCI < HC; N100(FCz-pure tone stimulus): MCI < HC; P200(FCz-pure tone stimulus): MCI < HC; P100(FCz-speech stimulus): MCI < HC; N100(FCz-speech	P100(FCz-pure tone stimulus): MCI > HC; N100(FCz-pure tone stimulus): MCI > HC; P200(FCz-pure tone stimulus): MCI > HC; P100(FCz-speech stimulus): MCI > HC; N100(FCz-speech	17
	13									
	17	72.80 (6.80)	8/9	15.90 (2.60)	/					

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Table 1 (continued)

Study	n	Age (SD)	Sex (M/F)	Education (SD)	MMSE (SD)	Task/Paradigm	ERP component and window (in ms post-event)	Results		JBI PACES
								Amplitude	Latency	
								stimulus): MCI < HC; P200(FCz-speech stimulus): MCI < HC	stimulus): MCI > HC P200(FCz-speech stimulus): MCI > HC	
(Papadaniil et al., 2016)(Papadaniil et al., 2016)	MCI 21 HC 21	72.00 (4.70) 67.00 (2.70)	7/14 8/13	10.00 (5.00) 14.00 (5.00)	27.00 (1.40) 28.81 (0.90)	Auditory Odd-ball Paradigm	MMN (Pz) 120–340 ms; P300 (Pz) 280–680 ms.	P300-Pz: MCI < HC; MMN-Pz: MCI > HC	P300-Pz: MCI > HC HC MMN-Pz: MCI > HC	18
(Papaliagkas et al., 2008)(V. Papaliagkas et al., 2008)	MCI 91 HC 30	66.60 (5.40) 68.90 (9.90)	35/ 56 15/ 15	/	/	Auditory Odd-ball Paradigm	N200 (Cz and Pz); P300 (Cz and Pz); Slow Wave (Cz and Pz).	N200: MCI > HC; P300: MCI > HC.	N200: MCI > HC; P300: MCI > HC; SW: MCI > HC.	16
(Papaliagkas et al., 2011)(V. T. Papaliagkas et al., 2011)	MCI 22 HC 30	67.40 (7.80) /	/	/	27.90 (1.90) /	Auditory Odd-ball Paradigm	N200 (Cz and Pz) 100–250 ms; P300 (Cz and Pz); Slow Wave (Cz and Pz).	N200: MCI > HC; P300: MCI > HC.	N200: MCI > HC; P300: MCI > HC; SW: MCI < HC.	15
(Gozke et al., 2013)(Gozke et al., 2013)	MCI 20 HC 20	66.00 (3.60) 66.30 (4.40)	12/8 14/6	/	/	Visual Odd-ball Paradigm	N200 (Fz, Cz, and Pz); P300 (Fz, Cz, and Pz).	N200: MCI > HC; P300: MCI < HC	N200: MCI > HC; P300: MCI > HC	13
(Lindín et al., 2013)(Lindín et al., 2013)	MCI 26 HC 30	67.80 (9.30) 63.90 (8.40)	12/ 14 9/21	10.15 (4.70) 9.4 (4.40)	25.8 (2.10) 28.5 (0.90)	Visual Odd-ball Paradigm	MMN (Cz) 125–260 ms.	MMN(Cz-novel minus standard): MCI > HC; MMN(Cz-deviant minus standard): MCI < HC	MMN(Cz-novel minus standard): MCI < HC; MMN(Cz-deviant minus standard): MCI < HC	17
(Stothart et al., 2015)(Stothart et al., 2015)	MCI 25 HC 26	77.30 (7.40) 76.00 (7.00)	16/9 14/ 12	/	25.90 (1.90) 28.50 (1.20)	Visual Odd-ball Paradigm	P100 (O1, Oz, O2, PO9, PO10, PO7, PO8) 75–170 ms; N100 (O1, Oz, O2, PO9, PO10, PO7, PO8) 160–260 ms; P300 (O1, Oz, O2, PO9, PO10, PO7, PO8) 230–500 ms; vMMN (O1, Oz, O2, PO9, PO10, PO7, PO8).	P100-Standard stimuli: MCI < HC; P100-Deviant stimuli: MCI < HC P100-Target stimuli: MCI < HC N100-Standard stimuli: MCI > HC N100-Deviant stimuli: MCI > HC N100-Target stimuli: MCI > HC P300-Target stimuli: MCI > HC	P100-Standard stimuli: MCI > HC; P100-Deviant stimuli: MCI > HC; P100-Target stimuli: MCI > HC; N100-Standard stimuli: MCI > HC; N100-Deviant stimuli: HC > MCI; N100-Target stimuli: MCI < HC; P300: P300-Target stimuli: MCI < HC	18
(Cespón et al., 2013)(Cespón et al., 2013)	MCI 30 HC 25	68.73 (9.21) 65.20 (8.20)	/	9.03 (4.45) 10.80 (5.60)	5.60 (2.58) 28.40 (1.30)	A Simon Task	N2pc (posterior contralateral) 200–300 ms; LRP.	P2pc(PO7/PO8-cdcp): HC < MCI P2pc(PO7/PO8-idcp): HC < MCI P2pc(PO7/PO8-cdip): HC < MCI P2pc(PO7/PO8-idip): HC < MCI LRP(C3/C4-cdcp): HC < MCI LRP(C3/C4-idcp): HC < MCI LRP(C3/C4-cdip): HC < MCI LRP(C3/C4-idip): HC < MCI	P2pc(PO7/PO8-cdcp): HC < MCI P2pc(PO7/PO8-idcp): HC < MCI P2pc(PO7/PO8-cdip): HC < MCI P2pc(PO7/PO8-idip): HC < MCI LRP(C3/C4-cdcp): HC > MCI LRP(C3/C4-idcp): HC > MCI LRP(C3/C4-cdip): HC > MCI LRP(C3/C4-idip): HC > MCI	15
(Cespón et al., 2015b)(Cespón et al., 2015b)	MCI 38 HC 15	67.65 (9.17) 66.00 (7.50)	/	8.62 (4.23) 10.20 (5.40)	6.39 (2.38) 28.30 (1.20)	A Simon Task	N200 (Fz, Cz, Pz, Oz) 200–375 ms; P300 (Fz, Cz, Pz, Oz) 375–700 ms; N2pc (PO7/PO8) 250–350 ms; MFN (Fz); PP (Fz).	N200(Pz-cdcp): HC < MCI N200(Pz-idcp): HC < MCI N200(Pz-cdip): HC > MCI N200(Pz-idip): HC > MCI P300(Pz-cdcp): HC > MCI P300(Pz-idcp): HC > MCI P300(Pz-cdip): HC > MCI	N200(Pz-cdcp): HC < MCI N200(Pz-idcp): HC < MCI N200(Pz-cdip): HC > MCI N200(Pz-idip): HC < MCI P300(Pz-cdcp): HC < MCI P300(Pz-idcp): HC < MCI P300(Pz-cdip): HC	15

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Table 1 (continued)

Study	n	Age (SD)	Sex (M/ F)	Education (SD)	MMSE (SD)	Task/ Paradigm	ERP component and window (in ms post- event)	Results		JBI PACES
								Amplitude	Latency	
								P300(Pz-idip): HC > MCI N2pc(PO7/PO8-cdcp): HC < MCI N2pc(PO7/PO8-idcp): HC < MCI N2pc(PO7/PO8-cdip): HC < MCI N2pc(PO7/PO8-idip): HC < MCI PP(Fz-cdcp): HC < MCI PP(Fz-idcp): HC > MCI PP(Fz-cdip): HC > MCI PP(Fz-idip): HC > MCI MFN(Fz-cdcp): HC > MCI MFN(Fz-idcp): HC > MCI MFN(Fz-cdip): HC < MCI MFN(Fz-idip): HC > MCI	< MCI P300(Pz-idip): HC < MCI N2pc(PO7/PO8-cdcp): HC < MCI N2pc(PO7/PO8-idcp): HC < MCI N2pc(PO7/PO8-cdip): HC < MCI N2pc(PO7/PO8-idip): HC < MCI	
(Cespón et al., 2015a)(Cespón et al., 2015a)	MCI 25 HC 18	70.11 (2.27) 68.30 (1.70)	13/12 7/11	9.67 (1.54) 11.10 (1.24)	25.36 (1.71) 28.50 (0.40)	A Simon Task	N2pc (posterior contralateral) 200–400 ms; N2cc (central contralateral) 200–400 ms.	N2cc: HC < MCI N2pc: HC < MCI	N2cc: HC < MCI N2pc: HC < MCI	17
(Wang et al., 2013)(Wang et al., 2013)	HC 16 MCI 15	69.30 (1.80) 72.90 (1.90)	9/7 9/6	14.00 (0.90) 12.80 (0.90)	29.30 (0.50) 27.00 (0.50)	Flanker Task	N200 (F3,Fz,F4) 200–350 ms; P300 (P3,P4,Pz) 300–550 ms.	N200-Non: HC < MCI N200-Neu: HC < MCI N200-Con: HC < MCI N200-Incon: HC < MCI P300-Non: MCI < HC P300-Neu: HC > MCI P300-Con: HC > MCI P300-Incon: HC > MCI	N200-Non: MCI < HC N200-Neu: HC < MCI N200-Con: HC < MCI N200-Incon: HC < MCI P300-Non: HC > MCI P300-Neu: HC < MCI P300-Con: HC = MCI P300-Incon: HC < MCI	18
(Tsai et al., 2016)(Tsai et al., 2016)	MCI 30 HC 30	68.23 (5.25) 66.87 (4.38)	12/18 14/16	12.37 (2.85) 12.87 (2.60)	27.60 (1.99) 28.57 (1.17)	Task-Switching	P300 (Fz, Cz, Pz) 300–650 ms	P300-non-switching: MCI < HC P300-switching: MCI < HC	P300-pure: MCI > HC P300-non-switching: MCI > HC P300-switching: MCI > HC	18
(Li et al., 2016)(Li et al., 2016)	MCI 24 HC 22	69.27 (7.55) 69.17 (8.91)	15/9 14/8	13.33 (2.80) 14.18 (3.09)	26.41 (2.12) 28.95 (0.95)	Delayed Matching to Sample (DMS) Task	N100 (P7,P8) 150–210 ms; P100 (O1, O2, P7, P8) 84–140 ms; N200 (F3, F4, C3, C4, CZ, FZ, FCz) 230–300 ms; P200 (FC1,CZ, FZ, FCz) 150–250 ms; P300 (O1,Pz) 250–450 ms.	P300(encoding epoch): MCI < HC; P2(encoding epoch): MCI < HC; N2(encoding epoch): MCI < HC; P1(encoding epoch): MCI < HC; N1(encoding epoch): MCI > HC; P300(retrieval epoch): MCI < HC; P2(retrieval epoch): MCI < HC; N2(retrieval epoch): MCI < HC; P1(retrieval epoch): MCI > HC; N1(retrieval epoch): MCI > HC.	P300(encoding epoch): MCI > HC; P2(encoding epoch): MCI > HC; N2(encoding epoch): MCI > HC; P1(encoding epoch): MCI < HC; N1(encoding epoch): MCI < HC; P300(retrieval epoch): MCI < HC; P2(retrieval epoch): MCI > HC; N2(retrieval epoch): MCI < HC; P1(retrieval epoch): MCI > HC; N1(retrieval epoch): MCI < HC.	18

Abbreviations: SD, standard deviation; JBI PACES, Joanna Briggs Institute-Practical Application of Clinical Evidence System; HC, healthy controls; MCI, mild cognitive impairment; M, male; F, female, ObA, object animal task; SiC: single car task; cdcp, compatible direction-compatible position; idcp, incompatible direction-compatible position; cdip, compatible direction-incompatible position; idip, incompatible direction-incompatible position; Con, congruent trials; Incon, incongruent trials; Neu, neutral; Non, none-target-alone.

We next analyzed the components with significant differences in the above meta-analysis by analyzing the paradigms as subgroups, and the resulting differences were all significant ($p < 0.05$). The detailed results are shown in Table 2.

3.6. Methodological quality assessment

The total JBI PACES scores for the studies included in this review are detailed in Table 1. Studies were scored on a scale of 1–20, with higher scores indicating higher qualities of evidence. Most studies scored 17 or 18, although 7 studies scored 15 or 16, and 2 studies scored 13. Overall, it appears that the literature included in the study is of high quality.

4. Discussion

Diamond proposed a multidimensional structural model of executive function, consisting of three core components: inhibitory control, working memory, and cognitive flexibility (Diamond, 2012). Based on this theoretical model, we selected experimental paradigms corresponding to the three core domains, and conducted a meta-analysis of the EEG components that are often analyzed by these paradigms. We aimed to perform a meta-analysis of executive function-related ERP studies in patients with MCI to explore potential electrophysiological biomarkers. Some components (such as P50, MFN, PP, ERS, and slow wave) were not included the meta-analysis because only a limited number of studies with similar designs reported their findings; however, most components were meta-analyzed (N100, N200, P100, P200, P300, and MMN). Our meta-analysis of the existing literature showed that when compared with HCs, patients with MCI showed reduced N200, P100, and P300 amplitudes, and extended P200, P300, and N200 latencies. However, MMN, N100, and P200 amplitudes as well as MMN and P100 incubation periods were not significantly different between patients with MCI and HCs.

4.1. P100

The P100 originates from the dorsal striatum and ventral striatal cortex, represents the processing of stimulus features and visuospatial selection (Hamilton et al., 2021), and is commonly found in the auditory/visual oddball paradigm and DMS task. The time window ranges from 80 to 130 ms, and the measurement indicators include latency and peak. The P100 is a positive wave that appears 100 ms after the presentation of the stimulus, and is related to the occipital facial area (Maestrini et al., 2023). Although studies on HC versus MCI groups have reported conflicting results regarding P100 composition, our meta-analysis suggests that the P100 amplitude is reduced in patients with MCI when compared with that in HCs, which is consistent with the findings of Stothart et al. (Stothart et al., 2015). The P100 amplitude in patients with AD is reduced compared with that in patients with MCI (Zhao et al., 2020). Considering these findings combined with our own, we speculate that the P100 amplitude shows a downward trend as the patient progresses to AD. Thus, P100 amplitude can discriminate between groups, thereby reflecting underlying neural deficits in patients with MCI.

4.2. P200

The P200 is associated with selective attentional or stimulus coding processes (Crowley and Colrain, 2004), and is commonly found in the auditory oddball and N-back paradigms. The P200 is an endogenous ERP component beginning at 180 ms, reaching its peak at 200 ms, ending at

350 ms, and mainly distributed in the middle frontal and parietal occipital lobes (Liu et al., 2021). Missonnier et al. noted that the P200 latency is delayed in patients with MCI compared with that in HCs. Our meta-analysis yielded the same results, demonstrating that P200 latency predicts MCI progression (Missonnier et al., 2007). These results revealed that attention, response selection, and execution were deficient in patients with MCI, and may also predict an important role for P100 amplitude and P200 latency in identifying patients with MCI, monitoring disease progression, and predicting disease prognosis.

4.3. P300

The P300 is mainly distributed in the frontal lobe electrode, which is related to automatic attention processing, novel stimuli, working memory updates, short-term memory, inhibitory control, and selective attentional processes (Patel and Azzam, 2005). It is caused by working memory updates, and is mainly distributed in the parietal lobe electrode (Polich, 2007). The P300 component is a positive-going wave that peaks between 300 and 500 ms post-stimulus and is commonly found in the go/no-go, N-back, and oddball paradigms. Its component can be broken down into two subcomponents: P300a and P300b. The studies including P300 component results included in this review did not classify them into P300a and P300b subcomponents, thus, the analysis and discussion process could not be divided by subcomponents. Our study showed that the peak value of P300 decreases as the incubation period increases, and is consistent with the findings of recent meta-analyses (Golob et al., 2002; Gozke et al., 2013). This study showed that compared with those in the HC group, the N200 and P300 latencies of patients with AD and MCI were prolonged (Howe, 2014; Howe et al., 2014). This finding may indicate the impairment of related functions in patients with AD and MCI.

4.4. N200

The N200 is associated with the decision-making process related to reaction inhibition (such as pre-attentive stimulus evaluation), or conflict detection and reaction preparation (Dong et al., 2009; Randall and Smith, 2011). It is a negative wave that typically peaks within 100–300 ms after stimulation and is commonly found in the go/no-go and oddball paradigms (Patel and Azzam, 2005). The N200 component can be broken down into two subcomponents: N200a and N200b, although few studies have isolated N200a and N200b. Therefore, these subcomponents could not be meta-analyzed in our study. Previous studies have shown that the N200 amplitude is increased and the latency is prolonged in patients with MCI versus HCs, which is consistent with our research results (Cespón et al., 2015a, 2015b; López Zunini et al., 2016a, 2016b; Wang et al., 2013). The N200 and P300 latencies are potentially accurate and sensitive indicators for predicting the progression of MCI (Bennys et al., 2007; Jiang et al., 2015).

Admittedly, some of the literature results are not in line with the present study, which may be due to differences in the experimental paradigm and the electrodes used for the analyses. If the analyses were performed strictly according to the same experimental paradigm and electrode classification, a meta-analysis could not be performed due to the small number of publications in the literature. Therefore, this study included all paradigms and electrodes containing the executive function of the same component in the analysis. Further and more rigorous meta-analyses should be performed in the future when the research in this field is more abundant.

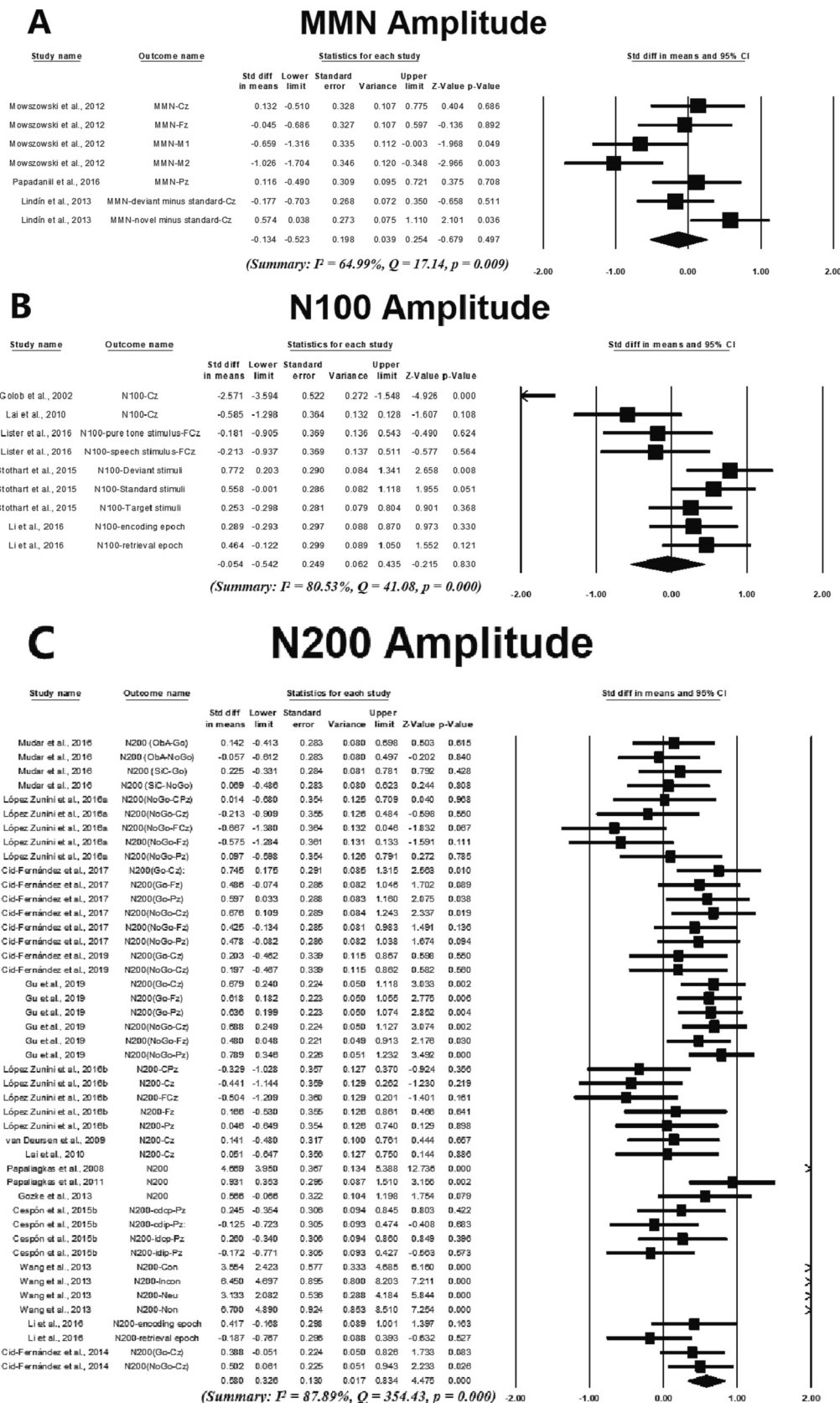


Fig. 2. Forest plot of MMN, N100, N200, P100, P200, and P300 amplitudes in MCI-related ERP studies. Diamonds denote pooled SMDs and 95 % CIs; positive values indicate increased amplitudes in patients with MCI, whereas negative values indicate increased amplitudes in HCs. Abbreviations: CI, confidence interval; ERP, event-related potential; HCs, healthy controls; MCI, mild cognitive impairment; SMD, standard mean difference; ObA, object animal task; SiC: single car task; cdcp, compatible direction-compatible position; idcp, incompatible direction-compatible position; cdip, compatible direction-incompatible position; idip, incompatible direction-incompatible position; Con, congruent trials; Incon, incongruent trials; Neu, neutral; Non, none-target-alone.

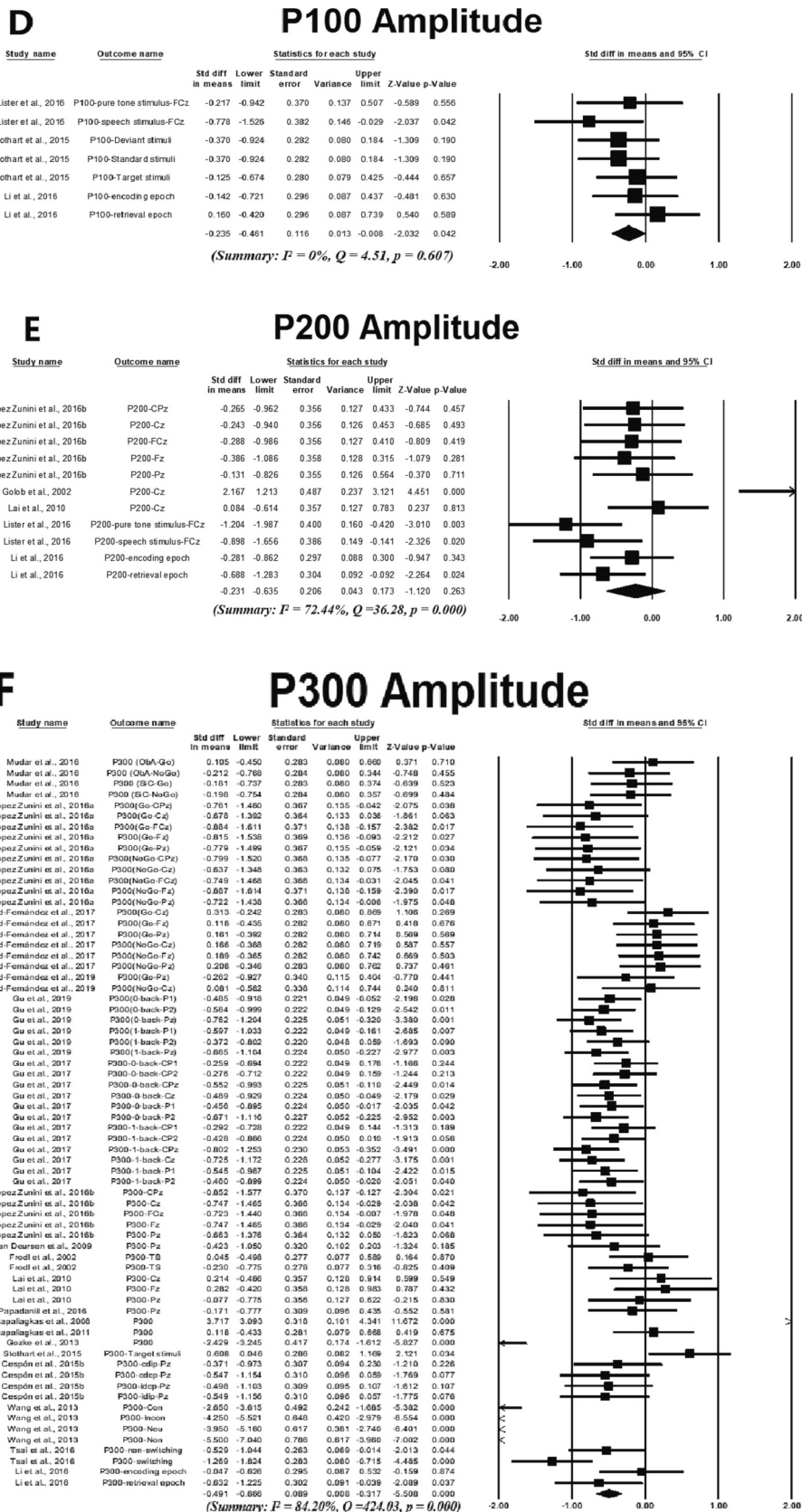


Fig. 2. (continued).

4.5. Heterogeneity

We observed high heterogeneity among studies of certain

components, such as the N100 and N200 component amplitudes, for several reasons. First, in this paper, meta-regression analyses were performed, and we found that differences in age, sex, and education

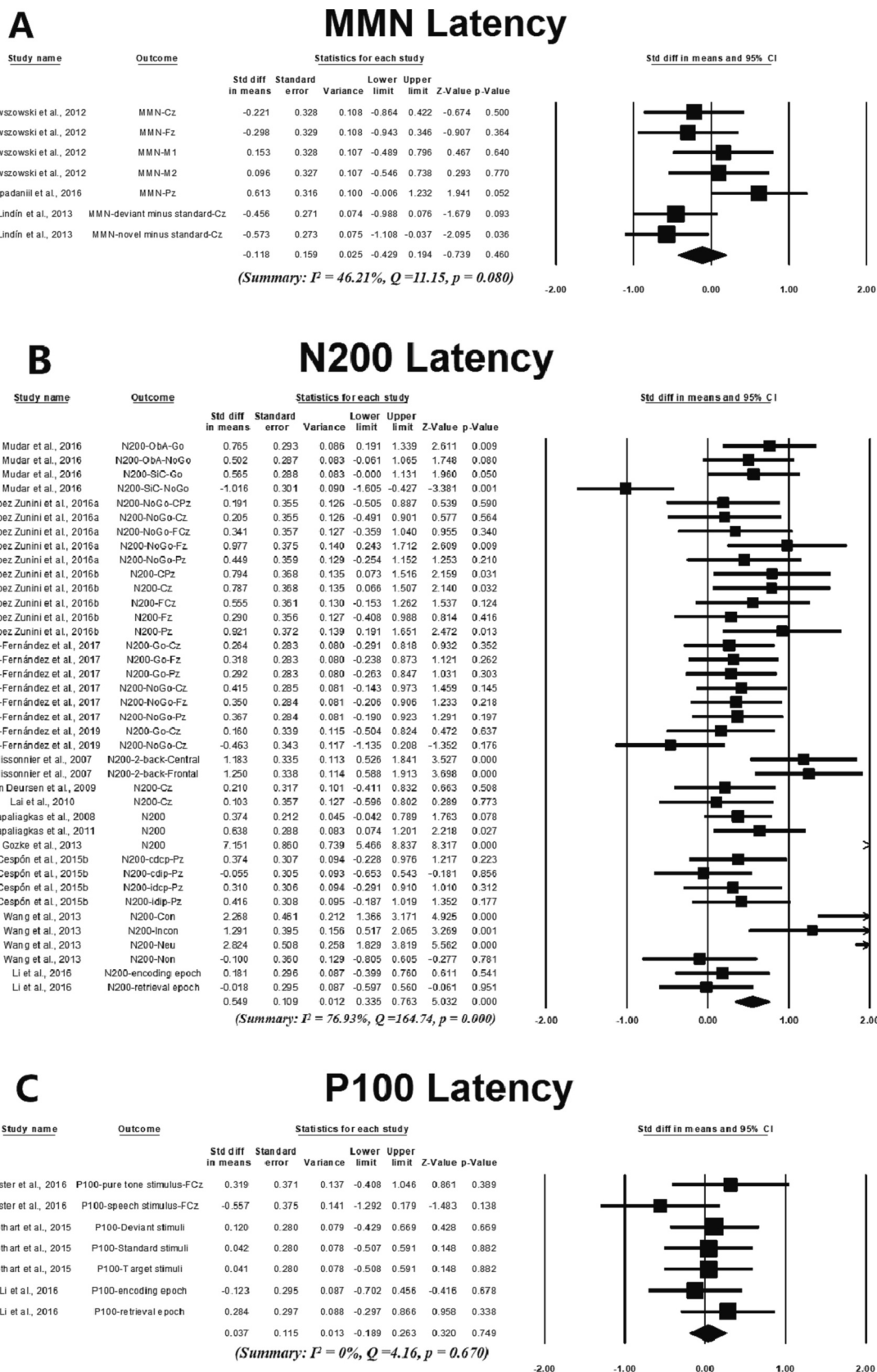


Fig. 3. Forest plot of MMN, N100, N200, P100, P200, and P300 latencies in MCI-related ERP studies. Diamonds denote pooled SMDs and 95 % CIs; positive values indicate increased amplitudes in patients with MCI, whereas negative values indicate increased amplitudes in HCs; Abbreviations: CI, confidence interval; ERP, event-related potential; HC, healthy controls; MCI, mild cognitive impairment; SMD, standard mean difference; ObA, object animal task; SiC: single car task; cdcp, compatible direction-compatible position; idcp, incompatible direction-compatible position; cdip, compatible direction-incompatible position; idip, incompatible direction-incompatible position; Con, congruent trials; Incon, incongruent trials; Neu, neutral; Non, none-target-alone.

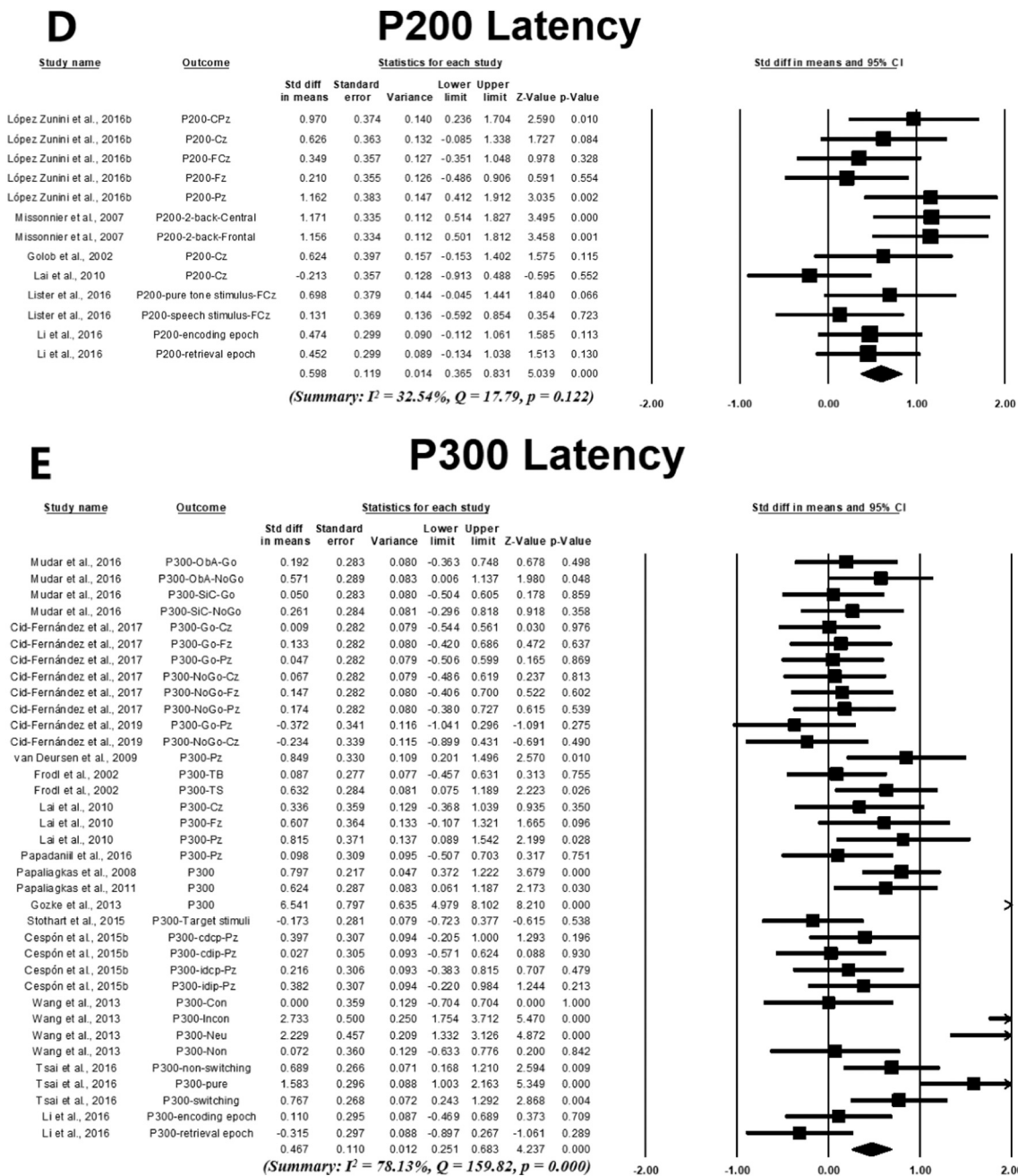


Fig. 3. (continued).

level could have contributed to the heterogeneity, consistent with previous studies (Polich and Corey-Bloom, 2005). In addition, most investigators rely on clinical assessments and cognitive tests in the diagnosis of MCI, factors that may partly contribute to these heterogeneities. Finally, our meta-analysis included various paradigms that were analyzed as subgroups in this study and differed significantly across components, suggesting that the differences in experimental paradigms may be another source of heterogeneity. Furthermore, controlling for factors such as disease severity and duration can reduce heterogeneity; however, achieving this can be difficult because studies reporting on MCI severity and duration are scarce.

4.6. Limitations

Compared with previous studies, this paper focuses more on EEG time-domain analyses of executive functions and achieves a higher

precision by narrowing the study population to older adults with MCI. However, this study had some limitations. First, for some of the sub-components, such as P50 and N2pc, the sample size was too small to conduct a meta-analysis. Follow-up studies should be conducted once research containing these components is abundant. Second, owing to the limited number of studies in the literature reporting on the related components, the individual components analyzed in this study, such as MMN, also have certain limitations. To overcome the limitations proposed in individual studies, more data are needed to reach a final conclusion. Finally, we only conducted a meta-analysis on MCI and did not search and analyze the various stages of MCI and AD from the perspective of disease progression. In the future, we will distinguish between the different stages of disease development and conduct meta-analyses to identify the most sensitive EEG physiological markers at different stages, thereby improving the process of disease diagnosis.

Table 2
Subgroup analysis results.

Components	Task/Paradigm	M ± SD	I ²	Q _w	Q _b	df	Z	p
N200 Amplitude	Simon	0.052 ± 0.153	0	0	175.103	5	3.934	0.000***
	Auditory	1.266 ± 0.792	96.541	115.625				
	DMS	0.112 ± 0.210	0	0				
	flanker	4.210 ± 0.335	0	0				
	Go/Nogo	0.297 ± 0.145	84.329	31.905				
P100 Amplitude	N-back	−0.210 ± 0.523	0	0	2.373	2	−2.032	0.042*
	Auditory	−0.488 ± 0.266	0	0				
	DMS	0.009 ± 0.209	0	0				
	Visual	−0.287 ± 0.163	0	0				
P300 Amplitude	Simon	−0.491 ± 0.154	0	0	194.299	7	−12.891	0.000***
	Auditory	0.541 ± 0.537	96.071	127.255				
	DMS	−0.332 ± 0.211	0	0				
	flanker	−3.751 ± 0.305	0	0				
	Go/Nogo	−0.202 ± 0.246	91.660	35.972				
	N-back	−0.546 ± 0.055	10.415	2.233				
	Task-switching	−0.872 ± 0.193	0	0				
	Visual	−0.895 ± 1.518	97.225	36.039				
	Simon	0.260 ± 0.153	0	0				
	Auditory	0.363 ± 0.138	0	1.680				
N200 Latency	DMS	0.081 ± 0.209	0	0	95.183	6	6.883	0.000***
	flanker	1.282 ± 0.210	0	0				
	Go/Nogo	0.264 ± 0.093	30.036	4.288				
	N-back	0.912 ± 0.276	72.872	3.686				
	Visual	7.151 ± 0.860	0	0				
P200 Latency	Auditory	0.277 ± 0.231	30.638	2.883	5.714	2	3.824	0.000***
	DMS	0.463 ± 0.211	0	0				
	N-back	0.875 ± 0.259	69.432	3.271				
P300 Latency	Simon	0.255 ± 0.153	0	0	40.852	6	7.474	0.000***
	Auditory	0.549 ± 0.103	6.351	5.339				
	DMS	−0.101 ± 0.209	0	0				
	flanker	0.913 ± 0.203	0	0				
	Go/Nogo	0.078 ± 0.130	51.875	4.156				
	Task-switching	0.975 ± 0.159	0	0				
	Visual	3.143 ± 3.356	98.417	63.166				

Note: Random-effects model was used; Q_w denotes within-group heterogeneity, Q_b denotes between-group heterogeneity; ****p* < 0.001, **p* < 0.05, same below.

5. Conclusions

EEG components of executive function may represent a promising approach to improving the identification of patients with MCI, monitoring disease progression, and predicting disease prognosis. In summary, our findings suggest that some ERP components such as N200, P100, and P300 amplitudes and P200, P300, and N200 latencies may be potential electrophysiological biomarkers for diagnosing MCI. Nevertheless, our study did have limitations, and therefore additional rigorous studies are needed on this topic. Further studies with the inclusion of more components will be conducted in the coming years, and these issues will be re-examined as more data become available.

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CRedit authorship contribution statement

Jiajun Che: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Nan Cheng:** Data curation, Investigation, Writing – original draft. **Bicong Jiang:** Formal analysis,

Investigation, Writing – review & editing. **Yanli Liu:** Software, Supervision, Writing – review & editing. **Yutong Li:** Resources, Software, Supervision. **Haining Liu:** Resources, Software, Supervision, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that there are no financial or other relationships that might lead to a conflict of interest regarding the present article. All authors have reviewed the final version of the manuscript and have approved it for publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijpsycho.2023.112295>.

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